

the entities with two inputs and one output each (Fig. 2). The larger block is the third entity with three inputs and one output. The component describes the interface of an entity that will be used as an instance (or a sub-block). A component instance is a distinct copy of the component that describes detail function of the signals associated with the component and is represented by a label in the code.

Program design

For a VHDL program, the user has to declare to the compiler the use of external files and libraries using the declaration statements like in C language. IEEE has created the basic VHDL library called IEEE.std_logic_1164, which declares different types of logic that will be used in the code. IEEE standards logic is included in most of the VHDL programs. A comment is prefixed by two hyphens, which are ignored by the VHDL compiler.

Each of the signals used in the TDM are generated internally. You can use the concept of a component and component instantiation to generate different signals from a single unit.

A single 'sig_genr' signal component is used for generating four input multiplexing signals s1, s2, s3, s4 and a master clock signal 'mclk'. The entity 'sig_genr' also represents the external interface in the code whereas the architecture is described by the 'arch_comp' section of the code. An if-condition in the code ensures that whenever enable input(en) is '1', the required signal is generated.

```
-- signal generator
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.ALL;

entity sig_genr is
generic (hperiod:time);
port(s1:inout std_logic;
en : in bit);
end entity sig_genr;
architecture arch_comp of sig_genr
```



Fig. 3: Project wizard on ISE Xilinx 9.2i

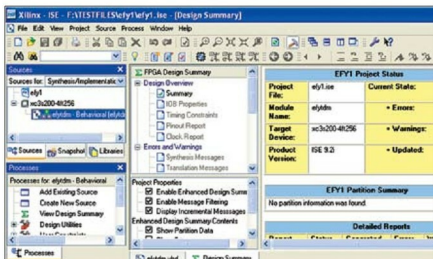


Fig. 4: Status of the EFY1 project wizard

```
is
begin
process
begin
if en = '1' then
s <= '1';
wait for hperiod;
s <= '0';
wait for hperiod;
end if;
end process;
end architecture arch_comp;
```

The entity name may be followed by the 'is' keyword and the 'end entity' keyword may be followed by the entity

name. The 'generic' keyword can be used for supplying the information when it is not a port signal in all the design units.

The component instantiation given below supplies half periods to get different signals.

```
sg1: sig_genr generic map (2ns)
port map (s1,'1');
sg2: sig_genr generic map (4ns)
port map (s2,'1');
sg3: sig_genr generic map (8ns)
port map (s3,'1');
sg4: sig_genr generic map (16ns)
port map (s4,'1');
```